

Data-Driven Networked Predictive Control for DoS-Attack-Resilient Remote Double-Tank Systems

1st Jianlei Gao

*China Industrial Control Systems
Cyber Emergency Response Team
Beijing, China
jianleixinye@163.com*

2nd Yun Li

*China Industrial Control Systems
Cyber Emergency Response Team
Beijing, China
18101260373@163.com*

3rd Yuchen Yang*

*China Industrial Control Systems
Cyber Emergency Response Team
Beijing, China
3694216334@163.com*

4th Mo Wang

*China Industrial Control Systems
Cyber Emergency Response Team
Beijing, China
wangmo0527@sina.com*

5th Shiwei Jia

*China Industrial Control Systems
Cyber Emergency Response Team
Beijing, China
2177500461@qq.com*

6th Tianyu Gong

*China Industrial Control Systems
Cyber Emergency Response Team
Beijing, China
125373679@qq.com*

Abstract—The rise of networked control systems (NCSs) has introduced new challenges in ensuring stability and performance under communication disruptions, especially denial-of-service (DoS) attacks. To tackle these issues, this paper introduces a novel data-driven predictive control framework tailored for NCSs, with a focus on the remote double-tank system as a benchmark. Leveraging a data-driven control methodology grounded in dynamic linearization and real-time input-output data, the proposed framework incorporates predictive mechanisms to handle packet losses, delays, and disorders induced by DoS attacks. The control scheme combines an adaptive parameter estimator, a predictive control generator, and a delay compensation module to maintain closed-loop stability under adverse network conditions. Simulation results confirm the efficacy of the proposed method in mitigating DoS-induced disruptions while sustaining superior control performance, underscoring its potential for reliable operation in uncertain and adversarial settings.

Index Terms—Networked control systems; predictive control; DoS attacks; data-driven control; remote double-tank systems.

I. INTRODUCTION

Network control systems (NCSs) have become a cornerstone of modern industrial automation, where control and monitoring are carried out over communication networks. These systems are widely deployed in applications such as process control, smart grids, and remote industrial plants [1] [2]. However, the reliance on network communication exposes NCSs to various vulnerabilities, including time delays, packet losses, and cyber-attacks. Among these threats, denial-of-service (DoS) attacks have drawn significant attention due to their ability to disrupt system performance by blocking communication channels [3].

The double-tank system, as a benchmark for process control, is commonly used to evaluate control strategies under network constraints [4]. Remote control of such systems faces significant challenges when exposed to cyber-attacks or network disturbances, such as time delays and DoS attacks. Effective control under such circumstances requires robust strategies that can mitigate the impact of network-induced issues while maintaining system performance.

Data-driven control (DDC), which bypasses the reliance on explicit mathematical models, has proven to be a viable substitute for conventional model-based approaches. By leveraging the system's input-output data directly, data-driven control ensures efficient system functionality despite uncertain dynamics or external disruptions [5] [6]. The flexibility and adaptability of data-driven methods make them particularly suitable for systems like the double-tank system, where obtaining accurate mathematical models can be challenging.

Recent studies have focused on improving the robustness of NCSs under DoS attacks. Strategies such as attack detection, communication scheduling, and resilient control design have shown potential in mitigating the effects of DoS attacks. However, many of these methods rely on accurate system models, which may not always be available in real-world applications. In contrast, data-driven control approaches, such as Model-Free Adaptive Control (MFAC) and Data-Driven Predictive Control (DDPC) [7] [8], have demonstrated their effectiveness in systems with limited or no explicit models. These methods utilize real-time system data to construct control laws and predict system behavior, making them highly adaptable to uncertain or changing environments.

Despite the progress in networked control and data-driven control methods, several challenges remain unaddressed. Most existing control strategies for NCSs rely on precise mathematical models, which are challenging to derive for complex systems such as the double-tank system. Additionally, while time-delay compensation methods have been well-studied, their applicability to DoS attacks is limited due to the unpredictable and intermittent nature of such attacks. Therefore, a unified framework that combines data-driven control with resilience to DoS attacks is urgently needed.

To tackle these issues, this study introduces an innovative data-driven predictive control framework tailored for networked control systems (NCSs) to counteract DoS attack effects, emphasizing its application on the remote double-tank system. The

principal contributions of this research are outlined as follows:

- 1) A data-driven control algorithm is designed to enhance robustness against DoS attacks without requiring an explicit system model.
- 2) A modified networked predictive control strategy is developed to handle both communication delays and intermittent packet losses caused by DoS attacks.
- 3) Simulation studies on the double-tank system denote the effectiveness of the proposed method in maintaining control performance under various attack scenarios.

The structure of this paper is outlined as follows. Section II covers the basic principles of model-free adaptive control and its application. Section III describes the proposed data-driven networked predictive control strategy, with an in-depth analysis of its implementation under DoS attack conditions. Section IV demonstrates the effectiveness of the proposed approach using simulations on a double-tank system. Lastly, Section V provides concluding remarks and suggests potential future research directions.

II. DATA-DRIVEN CONTROL

Data-driven control provides a versatile solution for discrete-time nonlinear systems by utilizing dynamic linearization and pseudo-partial derivatives (PPD) extracted from input-output data. It eliminates the need for explicit system models or prior knowledge, such as order or time delay, enabling adaptive control design directly from real-time data [9].

Compared to traditional methods, data-driven control simplifies implementation by removing the need for external test signals or high-order controllers, even in complex systems. Techniques like compact-form dynamic linearization (CFDL) [10] and its extensions streamline controller design and improve practicality. This makes data-driven control a cost-effective and robust solution for industrial applications under uncertain conditions.

A. Compact-Form Dynamic Linearization

Consider the following discrete-time nonlinear system with multiple inputs and outputs (MIMO):

$$\begin{aligned} \mathbf{y}((k+1)T) &= \Gamma(\mathbf{y}(kT), \dots, \mathbf{y}((k-n_y)T), \\ &\quad \mathbf{u}(kT), \dots, \mathbf{u}((k-n_u)T)). \end{aligned} \quad (1)$$

Here, T denotes the sampling interval, $\mathbf{y}(kT)$ and $\mathbf{u}(kT)$ stand for the output and input vectors at the sampling time kT , respectively. The parameters n_y and n_u are unknown positive integers, while $\Gamma(\cdot)$ is an unspecified nonlinear mapping.

Assumption 2.1: The partial derivatives of $\Gamma(\cdot)$ with respect to its $(n_y + 2)$ th argument are continuous, except at a finite set of time points.

Assumption 2.2: The system in (1) adheres to the generalized Lipschitz condition at all time points except for a finite few. Specifically, for any $k_1 > k_2 \geq 0$ and $\mathbf{u}(k_1T) \neq \mathbf{u}(k_2T)$, the following inequality is satisfied:

$$\|\mathbf{y}((k_1+1)T) - \mathbf{y}((k_2+1)T)\| \leq b\|\mathbf{u}(k_1T) - \mathbf{u}(k_2T)\|.$$

Here, $\mathbf{y}((k_i+1)T)$ is defined as $\Gamma(\mathbf{y}(k_iT), \dots, \mathbf{y}((k_i-n_y)T), \mathbf{u}(k_iT), \dots, \mathbf{u}((k_i-n_u)T))$, where $i = 1, 2$, and b is a positive constant.

To simplify analysis, we denote the output increment between consecutive steps as $\Delta\mathbf{y}((k+1)T) = \mathbf{y}((k+1)T) - \mathbf{y}(kT)$ and the input increment as $\Delta\mathbf{u}(kT) = \mathbf{u}(kT) - \mathbf{u}((k-1)T)$.

Theorem 2.1: Given a nonlinear system in (1) that fulfills Assumptions 2.1 and 2.2, if $\|\Delta\mathbf{u}(kT)\| \neq 0$, then there exists a time-varying pseudo-Jacobian matrix $\Psi_c(kT)$ such that the system can be equivalently represented in a compact-form dynamic linearization model:

$$\Delta\mathbf{y}((k+1)T) = \Psi_c(kT)\Delta\mathbf{u}(kT), \quad (2)$$

where

$$\Psi_c(kT) = \begin{bmatrix} \psi_{11}(kT) & \psi_{12}(kT) & \cdots & \psi_{1m}(kT) \\ \psi_{21}(kT) & \psi_{22}(kT) & \cdots & \psi_{2m}(kT) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{m1}(kT) & \psi_{m2}(kT) & \cdots & \psi_{mm}(kT) \end{bmatrix} \in \mathbb{R}^{m \times m}. \quad (3)$$

In this matrix, $\Psi_c(kT)$ is the pseudo-Jacobian matrix of the system. When $\Psi_c(kT)$ is a 1×1 matrix, it is referred to as the pseudo-partial derivative of the system.

The dynamic linearization model uses only input-output data from the closed-loop system, avoiding the need for structural or parametric details of system dynamics. By leveraging incremental input-output data, this method simplifies controller design while remaining unsuitable for detection or diagnosis tasks [11]. Unlike Taylor linearization, which retains higher-order terms, or feedback linearization, which requires precise system structures, dynamic linearization focuses on mapping input and output increments. Orthogonal function approximation, in contrast, often sacrifices control accuracy by projecting nonlinearities onto pre-defined bases.

B. Data-Driven Control Theory

Consider the following control input cost function:

$$\begin{aligned} J(\mathbf{u}(kT)) &= \|\mathbf{y}^*((k+1)T) - \mathbf{y}((k+1)T)\|^2 \\ &\quad + \lambda \|\mathbf{u}(kT) - \mathbf{u}((k-1)T)\|^2, \quad \lambda > 0. \end{aligned} \quad (4)$$

Here, $\lambda > 0$ denotes a weighting parameter to prevent significant variations in the control input. The term $\mathbf{y}^*((k+1)T)$ indicates the target output signal at time $(k+1)T$.

By applying equation (1) to the cost function (4) and simplifying the expression, the control algorithm is obtained for a single-input single-output (SISO) system:

$$\begin{aligned} \mathbf{u}(kT) &= \mathbf{u}((k-1)T) \\ &\quad + \frac{\rho \Psi_c^T(kT)}{\lambda + \|\Psi_c(kT)\|^2} (\mathbf{y}^*((k+1)T) - \mathbf{y}(kT)), \end{aligned} \quad (5)$$

where $\rho > 0$ denotes the step size, and λ represents a weighting factor aimed at ensuring smooth control inputs. This method aids in maintaining system stability while achieving satisfactory tracking performance.

The next step involves formulating an estimation algorithm for the pseudo-Jacobian matrix $\Psi_c(kT)$. Based on the compact-form dynamic linearization model for MIMO discrete-time systems, the parameter estimation cost function is defined as:

$$J(\Psi_c(kT)) = \|\Delta \mathbf{y}(kT) - \Psi_c(kT)\Delta \mathbf{u}((k-1)T)\|^2 + \mu \|\Psi_c(kT) - \hat{\Psi}_c((k-1)T)\|^2, \quad \mu > 0. \quad (6)$$

where $\mu > 0$ serves as a weighting parameter to limit large variations in $\hat{\Psi}_c$, the estimated pseudo-Jacobian matrix.

Minimizing the cost function (6) leads to a simplified pseudo-partial derivative approach for estimating $\hat{\Psi}_c$, which avoids the complexity of matrix inversion for high-dimensional control inputs and outputs. The final data-driven control framework for discrete-time nonlinear MIMO systems, integrating parameter estimation and control algorithms, is presented as:

$$\begin{aligned} \hat{\Psi}_c(kT) &= \hat{\Psi}_c((k-1)T) \\ &+ \eta \frac{[\Delta \mathbf{y}(kT) - \hat{\Psi}_c((k-1)T)\Delta \mathbf{u}((k-1)T)] \Delta \mathbf{u}^T((k-1)T)}{\mu + \|\Delta \mathbf{u}((k-1)T)\|^2}, \end{aligned} \quad (7)$$

$$u(kT) = u((k-1)T) + \frac{\rho \hat{\Psi}_c^T(kT)}{\lambda + \|\hat{\Psi}_c(kT)\|^2} [y^*((k+1)T) - y(kT)], \quad (8)$$

where $\hat{\psi}_{ij}(1T)$ is the initial value of $\hat{\psi}_{ij}(kT)$, $i, j = 1, \dots, m$, $i \neq j$, and the following conditions apply:

- For $i = 1, \dots, m$:

$$\begin{aligned} \hat{\psi}_{ii}(kT) &= \hat{\psi}_{ii}(1T), \\ &\text{if } |\hat{\psi}_{ii}(kT)| \geq b_2 \text{ or } |\hat{\psi}_{ii}(kT)| \leq \alpha b_2, \\ &\text{or } \text{sign}(\hat{\psi}_{ii}(kT)) \neq \text{sign}(\hat{\psi}_{ii}(1T)). \end{aligned}$$

- For $i, j = 1, \dots, m$, $i \neq j$:

$$\begin{aligned} \hat{\psi}_{ij}(kT) &= \hat{\psi}_{ij}(1T), \\ &\text{if } |\hat{\psi}_{ij}(kT)| \leq b_1 \\ &\text{or } \text{sign}(\hat{\psi}_{ij}(kT)) \neq \text{sign}(\hat{\psi}_{ij}(1T)). \end{aligned}$$

where $\lambda > 0$, $\eta \in (0, 2]$ is the step size factor, ρ are positive constants, and $\Delta \mathbf{u}((k-1)T)$ represents the input increment.

This approach ensures robust performance by iteratively updating the pseudo-Jacobian matrix and computing control inputs utilizing real-time input and output information. The data-driven control scheme integrates a parameter identification algorithm with a control law that dynamically adapts to changes in the system, maintaining stability and improving tracking accuracy under uncertain conditions. At each time step, the estimated $\hat{\Psi}_c(kT)$ is used to update the control law, and the resulting control action generates new data for subsequent iterations, ensuring continuous adaptation and robustness.

Assumption 2.3: Given a bounded reference signal $y^*((k+1)T)$, a bounded control input $u^*(kT)$ can be found such that the system output matches $y^*((k+1)T)$.

Assumption 2.4: For $\Delta u(kT) \neq 0$, the pseudo-Jacobian matrix $\Psi_c(kT)$ retains the invariance of its two-norm sign, i.e., $\hat{\psi}_{ij}(kT) > \epsilon > 0$, or $\hat{\psi}_{ij}(kT) = -\epsilon$, where ϵ represents a minor positive constant.

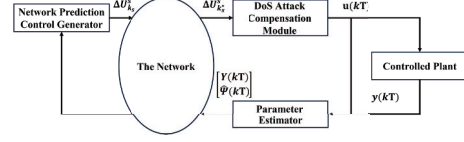


Fig. 1: Networked Predictive Control System Architecture

While data-driven control offers adaptability and robustness, it may not outperform model-based methods in all scenarios. However, for discrete-time nonlinear systems satisfying Assumptions 2.1-2.4, it ensures output controllability (Assumption 2.3) and quasi-linear behavior (Assumption 2.4), where increasing input does not reduce output. This makes it suitable for systems like temperature regulation and remote double-tank setups, where it demonstrates broad applicability.

III. DATA-DRIVEN NETWORKED PREDICTIVE CONTROL UNDER DoS ATTACKS

Direct utilization of data-driven control in networked control systems can result in instability or oscillations due to communication-related problems. To overcome issues such as packet drops, disordered data, and delays induced by DoS attacks, this study incorporates predictive control [12] for formulating a data-driven network control framework.

Assumption 3.1: Sensors and actuators operate in a time-driven manner with synchronized clocks, whereas controllers follow an event-driven approach.

Assumption 3.2: The random loop delay τ_k is bounded, i.e., $\tau_k \leq \bar{\tau}$.

In networked environments, loop delays from DoS attacks include transmission and processing times. Lost packets are treated as having infinite delays, and disordered packets have larger delays than ordered ones. To ensure real-time control, only packets reaching the actuator are used, with disordered packets discarded. The loop delay is redefined as the time difference between the actuator's most recently received packet and the current time, remaining bounded even under DoS attacks, thus capturing the effects of packet disorder and loss.

From the nonlinear discrete-time system assumptions in Section 2, the dynamic parameter update and control increment are derived as follows:

$$\begin{aligned} \hat{\Psi}_c(kT) &= \hat{\Psi}_c((k-1)T) \\ &+ \eta \frac{[\Delta \mathbf{y}(kT) - \hat{\Psi}_c((k-1)T)\Delta \mathbf{u}((k-1)T)] \Delta \mathbf{u}^T((k-1)T)}{\mu + \|\Delta \mathbf{u}((k-1)T)\|^2}, \end{aligned} \quad (9)$$

$$\begin{aligned} \Delta \mathbf{u}(kT) &= (\lambda I + \Psi_c^T(kT)\Psi_c(kT))^{-1} \\ &\times \rho \Psi_c^T(kT) (y^*((k+1)T) - \mathbf{y}(kT)). \end{aligned} \quad (10)$$

where $\hat{\Psi}_c(kT)$ is the estimated pseudo-Jacobian matrix, $y^*((k+1)T)$ denotes the reference output, while $\lambda > 0$, $\eta \in (0, 1)$, $\mu > 0$, and $\rho > 0$ represent the system parameters.

A data-driven predictive control framework is presented to ensure closed-loop stability under DoS attacks. As depicted in Fig. 1, it comprises three components: a parameter estimator,

a network prediction control generator, and a DoS attack compensation module.

A. Parameter Estimator

The parameter estimator calculates the dynamic linearization parameter vector $\Psi_c(kT)$ using the system's input-output data. Simultaneously, the sensor embeds the system output at time kT into the data packet D_k :

$$D_k = [\mathbf{y}(kT), \Psi_c(kT)]^T. \quad (11)$$

B. Network Prediction Control Generator

Given feedback from the parameter estimator, the predicted control increment is computed as:

$$\Delta \mathbf{u}((k_s)T|k_sT) = \alpha(k_sT) [\mathbf{y}^*((k_s + 1)T) - \mathbf{y}(k_sT)], \quad (12)$$

where $\alpha(k_sT) = \frac{\rho \Psi_c^T(k_sT)}{\lambda + \|\Psi_c(k_sT)\|^2}$ and k_sT is the timestamp of the most recent packet. The predicted control and output variables over the horizon can be expressed as:

$$\Delta \mathbf{y}((k_s + i)T|k_sT) = \Psi_c(k_sT) \Delta \mathbf{u}((k_s + i - 1)T|k_sT), \quad (13)$$

$$\begin{aligned} \mathbf{y}((k_s + i)T|k_sT) &= \mathbf{y}((k_s + i - 1)T|k_sT) \\ &\quad + \Delta \mathbf{y}((k_s + i)T|k_sT), \end{aligned} \quad (14)$$

$$\begin{aligned} \Delta \mathbf{u}((k_s + i)T|k_sT) &= \alpha(k_sT) [\mathbf{y}^*((k_s + i + 1)T) \\ &\quad - \mathbf{y}((k_s + i)T|k_sT)]. \end{aligned} \quad (15)$$

The total control increment $\Delta \mathbf{u}_s$ is:

$$\begin{aligned} \Delta \mathbf{u}_s((k_s + i)T|k_sT) &= \Delta \mathbf{u}_s((k_s + i - 1)T|k_sT) \\ &\quad + \Delta \mathbf{u}((k_s + i)T|k_sT). \end{aligned} \quad (16)$$

C. DoS Attack Compensation Module

Assuming the last usable data packet has a timestamp k_s^* ($k_s^* \leq k_s$), the packet contains:

$$\begin{aligned} \Delta \mathbf{U}_{s, k_s^*} &= [\Delta \mathbf{u}_s(k_s^*T|k_s^*T), \Delta \mathbf{u}_s((k_s^* + 1)T|k_s^*), \\ &\quad \dots, \Delta \mathbf{u}_s((k_s^* + \bar{\tau})T|k_s^*T)]^T. \end{aligned} \quad (17)$$

The loop delay caused by DoS attacks is:

$$\tau_k = (k - k_s^*)T. \quad (18)$$

The final control output compensates for delay-induced errors:

$$\mathbf{u}(kT) = \mathbf{u}((k_s^* - 1)T) + \Delta \mathbf{u}_s(kT|k_s^*T), \quad (19)$$

where

$$\Delta \mathbf{u}(kT) = \Delta \mathbf{u}(kT|k_s^*T). \quad (20)$$

IV. SIMULATION AND RESULTS ANALYSIS

To assess the performance of the proposed data-driven networked predictive control framework against denial-of-service (DoS) attacks, simulations are carried out on a double-tank system. This system serves as a well-established benchmark for evaluating control strategies under network conditions. In this research, a neural network is utilized to model the dynamics of the double-input double-output (DIDO) double-tank system. The constructed neural network model accounts for system uncertainties and nonlinear characteristics, providing a reliable platform to test the robustness of the proposed control approach.

A. Simulation Setup

The double-tank system comprises two linked tanks. The flow rates managed by pumps serve as the inputs, while the outputs correspond to the water levels in the upper and lower tanks. A neural network model is trained using input-output data collected under various operating conditions, ensuring that it accurately represents the system's non-linear behavior.

For the networked control scenario, the following types of DoS attacks are simulated:

- **Packet loss:** Random packet drops in both forward and feedback communication channels.
- **Packet disorder:** Out-of-sequence packet arrivals at the controller or actuator.
- **Induced delays:** Variable time delays in communication channels within a bounded range.

The control objective is to maintain the water levels at the desired setpoints while ensuring robustness against communication interruptions.

B. Simulation Results

The performance of our method is evaluated under two scenarios: normal data-driven control without prediction and data-driven networked predictive control under DoS attacks.

1) *Normal Data-Driven Control without Prediction:* In the first scenario, normal data-driven control without prediction is applied, simulating network-induced delays without DoS attacks. The outputs of the upper and lower tanks are shown in Fig. 2. As seen in the figures, the control method struggles to track the reference setpoints effectively due to time delays in the communication channels.

Fig. 3 shows the control inputs applied during the simulation. It is evident that the control input signals exhibit significant fluctuations, leading to poor tracking performance.

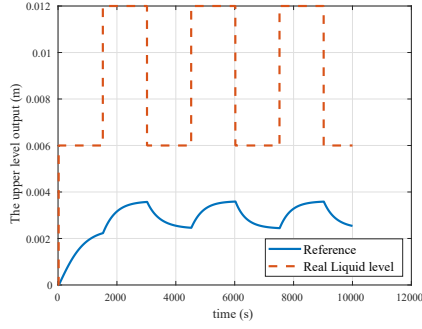
2) *Data-Driven Networked Predictive Control under DoS Attacks:* In the second scenario, networked predictive control driven by data is applied in simulated DoS attacks. The responses of the upper and lower tank levels are shown in Fig. 4. Compared to the first scenario, the predictive control method significantly improves the tracking performance, with both tank levels closely following their respective setpoints despite packet losses, disorder, and induced delays.

Fig. 5 shows the estimated pseudo-Jacobian matrix used in the dynamic linearization process. The results indicate smoother control actions and better stability compared to the scenario without predictive control.

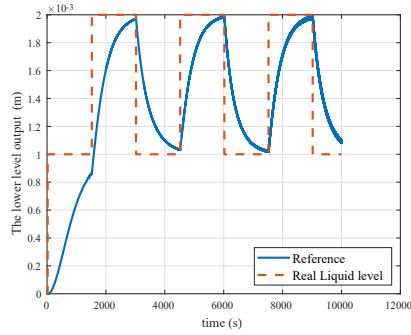
The simulation results demonstrate that the proposed data-driven networked predictive control framework is effective in mitigating the effects of DoS attacks, ensuring robust performance under various communication disruptions.

V. CONCLUSION

This work introduces a data-driven networked predictive control framework to tackle challenges posed by denial-of-service (DoS) attacks in networked control systems (NCSs). Instead of relying on explicit mathematical models, the approach leverages input-output data, simplifying control design and effectively



(a) Upper tank level



(b) Lower tank level

Fig. 2: Tank level outputs under normal data-driven control without prediction

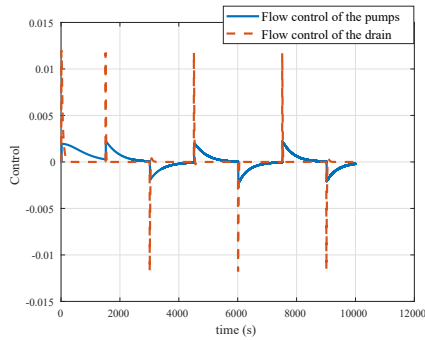
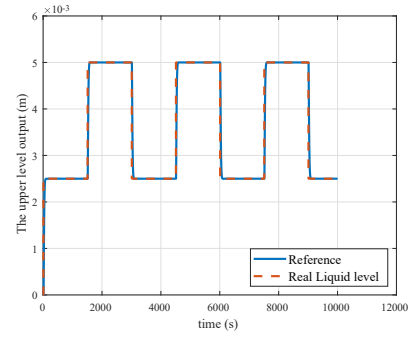


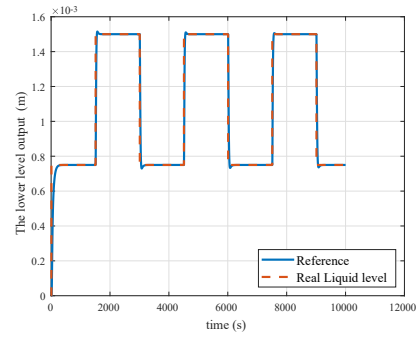
Fig. 3: Control input under normal data-driven control without prediction

mitigating communication issues such as packet loss, delay, and disorder. The proposed framework incorporates dynamic linearization-based data-driven control and predictive control mechanisms to bolster system robustness and sustain stable performance under unfavorable network conditions.

Simulation results on a remote double-tank system confirm the method's effectiveness in maintaining closed-loop stability and ensuring robust control performance across different DoS attack scenarios. These results highlight the framework potential for practical deployment in real-world industrial applications,



(a) Upper tank level with prediction



(b) Lower tank level with prediction

Fig. 4: Tank level outputs under data-driven networked predictive control

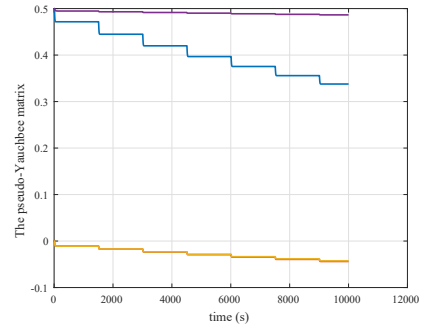


Fig. 5: Estimated pseudo-Jacobian matrix under data-driven networked predictive control

particularly in uncertain and demanding conditions.

Future research will extend the proposed framework to more complex systems and explore its application in multi-agent networked control scenarios. Additionally, experimental validation on real-world hardware setups will be conducted to further assess the framework's practical applicability and effectiveness.

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